

2(a)

any **four** from:

- 1** (in combination treatment) antibiotics (in Table 2.1), act at different targets / have different modes of action / AW ;  
**A** comparison of any two antibiotics from Table 2.1  
**A** suggestion of how two antibiotics have different ways of killing
- 2** *idea that* if resistance / mutation, occurs / exists, unlikely to be against all antibiotics / other antibiotics should still be effective ;
- 3** (in combination treatment) if resistance / mutation, occurs / exists, all bacteria will (still) be, killed / destroyed / AW ;  
**A** no bacteria remain to develop resistance / no reservoir of resistant bacteria
- 4** antibiotic resistance, not / less likely to be, spread to affect people because no bacteria surviving (with combination treatment) ; AW
- 5** long treatment time / 6 months, with, combination treatment / AW, increases chance of killing all bacteria  
**or**  
long treatment time with a single antibiotic not effective in killing all bacteria if, resistance develops / a mutation occurs ;
- 6** AVP ; e.g.  
combination treatment (is likely to) eliminate bacteria more quickly (so less chance of resistance occurring)  
  
resistance to different antibiotics involves more genes so less chance of resistance occurring  
  
gene for antibiotic resistance has more chance of being passed on if using single antibiotic (and not all killed) ora  
  
if using single antibiotic (and not all killed) more chance of being passed on (to other bacteria) by horizontal / vertical / AW, transmission ora

4

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(b)(i)  | transcription ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
| 2(b)(ii) | <p><i>any <b>three</b> from:</i></p> <p><b>I</b> stops transcription</p> <p><b>1</b> alters shape of / blocks, active site ;<br/> <b>R</b> enters / fits into, active site <i>as it is <u>not</u> competitive</i><br/> <b>R</b> <i>ref. to</i> choice of competitive or non-competitive<br/> <b>A</b> alters shape of enzyme <i>only if mp2 gained</i></p> <p><b>2</b> substrate / nucleotides, cannot, bind to / enter, active site ;<br/> <b>A</b> fewer / no, enzyme-substrate complexes / ESCs, form<br/> <i>allow ecf from mp1 if rifampicin described as, competitive / enters active site</i></p> <p><b>3</b> complementary (base) pairs / (complementary) base pairs, cannot form / form less easily (between DNA and RNA nucleotides) ;</p> <p><b>4, 5, 6</b> <i>(in context of rifampicin binding to RNA polymerase)</i><br/> prevents / AW, unwinding of DNA (double helix) ;<br/> <b>A</b> uncoiling<br/> <b>I</b> unzipping prevents / AW, attachment to DNA (strand)<br/> <b>A</b> prevents / AW, attachment to promoter<br/> prevents / AW, movement along DNA (strand) ;<br/> prevents / AW, elongation (of polynucleotide) / formation of polynucleotide / nucleotides being joined ;<br/> prevents / AW, phosphodiester bond formation ;<br/> prevents / AW, proof reading ;</p> <p><b>7</b> AVP ; e.g. prevents / AW, tRNA / rRNA, formation</p> | 3     |
| 2(c)(i)  | Phe<br>Tyr ; <i>one mark for both correct</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1     |
| 2(c)(ii) | <b>B D F G</b> ; <i>in any order</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(c)(iii) | <p><i>resistance</i></p> <p><b>1</b> shape / tertiary structure, of, <math>\beta</math>-subunit / enzyme, altered / AW ;<br/> <b>I</b> active site changes shape<br/> <b>A</b> quaternary structure of enzyme altered</p> <p><b>2</b> rifampicin / antibiotic, cannot / does not, bind (as well) ; AW<br/> <b>R</b> if context is binding to active site</p> <p><i>different levels of resistances</i><br/> any <b>two</b> from:</p> <p><b>3</b> (because) other mutations are involved ; AW<br/> <b>A</b> described from Table 2.1<br/> e.g. <b>C</b> has 2 (other) mutations and <b>D</b> and <b>E</b> have 3</p> <p><b>4</b> mutations may result in different changes to, structure / shape of,<br/> <math>\beta</math>-subunit / enzyme, and result in different, effects / levels of resistance ;</p> <p><b>5</b> data from Table 2.2 to support mp 4 ; <i>must be linked to concept of mp4</i></p> <p><b>6</b> AVP ; e.g. <i>ref. to</i> different binding abilities (of rifampicin to enzyme)<br/> some (of the other) mutations may cause more of a change to binding<br/> site for rifampicin<br/> mutation(s) may make it harder to bind<br/> lower resistance = binds, more strongly / for longer time<br/> lower resistance = higher proportion of transcription events hindered<br/> e.g. idea that mutations still, produce functioning enzyme / allow catalysis<br/> to occur</p> | <b>3</b> |